

Markscheme

Mathematics: analysis and approaches
Practice question bank

122. (a)

A committee of 4 is chosen from a group of 10 people, which includes two particular people, X and Y.

Find the number of ways of choosing the committee if X and Y cannot both be on the committee together.

total ways: $\binom{10}{4} = 210$ *(M1)* **A1**

ways with both X and Y included: $\binom{8}{2} = 28$ *(M1)*

subtract: $210 - 28$ *(M1)*

$= 182$ **A1**

[5 marks]